

BIO-3105: Chapter-7

Biological Data and Databases

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Nucleotide

Alphabet of Life

Search: Nucleotide [Limits](#) [Advanced search](#) [Help](#)[Display Settings:](#) ☒ GenBank

Dengue virus type 1, complete genome

NCBI Reference Sequence: NC_001477.1

[FASTA](#) [Graphics](#)[Go to:](#)

LOCUS	NC_001477	10735 bp ss-RNA	linear	VRL 08-DEC-2008
DEFINITION	Dengue virus type 1, complete genome.			
ACCESSION	NC_001477			
VERSION	NC_001477.1 GI:9626685			
DBLINK	Project: 15306			
KEYWORDS	.			
SOURCE	Dengue virus 1			
ORGANISM	Dengue virus 1			
	Viruses; ssRNA positive-strand viruses, no DNA stage; Flaviviridae; Flavivirus; Dengue virus group.			
REFERENCE	1 (bases 1 to 10735)			
AUTHORS	Puri,B., Nelson,W.M., Henchal,E.A., Hoke,C.H., Eckels,K.H., Dubois,D.R., Porter,K.R. and Hayes,C.G.			
TITLE	Molecular analysis of dengue virus attenuation after serial passage in primary dog kidney cells			
JOURNAL	J. Gen. Virol. 78 (PT 9), 2287-2291 (1997)			
PUBMED	9292016			

<https://www.ncbi.nlm.nih.gov/>

Introduction to Database

- A database is a **computerized archive** used to **store and organize** data in such a way that information can be retrieved easily via a variety of search criteria. Databases are composed of computer hardware and software for data management.
- Each **record**, also called **an entry**, should contain a number of fields that hold the actual data items, for example, fields for names, phone numbers, addresses, dates.
- To **retrieve** a particular record from the database, a user can specify a particular piece of information, called value, to be found in a particular field and expect the computer to retrieve the whole data record. This process is called **making a query**

Biological Databases

- Based on their contents, biological databases can be roughly divided into three categories: primary databases, secondary databases, and specialized databases.
- **Primary databases** contain original biological data. They are archives of raw sequence or structural data submitted by the scientific community. GenBank and Protein Data Bank (PDB) are examples of primary databases.
- **Secondary databases** contain computationally processed or manually curated information, based on original information from primary databases. Translated protein sequence databases containing functional annotation belong to this category. Examples are SWISS-Prot and Protein Information Resources (PIR)
- **Specialized databases** are those that cater to a particular research interest. For example, Flybase, HIV sequence database, and Ribosomal Database Project are databases that specialize in a particular organism or a particular type of data

Various Omics-study

Genomics: The study of the complete genome of an organism, including gene content, structure, and variation.

Proteomics: Large-scale examination of the entire set of proteins (proteome) produced by an organism, focusing on structures, functions, and interactions.

Transcriptomics: Analysis of the complete set of RNA molecules (transcriptome), such as mRNA, rRNA, tRNA, and non-coding RNAs, to assess gene expression patterns.

Metabolomics: Systematic study of small-molecule metabolites (metabolome) within cells, tissues, or organisms to capture chemical fingerprints of biological processes.

Relation between Multi-Omics and Bioinformatics

- Multi-omics approaches integrate data from multiple biological layers (genomics, transcriptomics, proteomics) to provide holistic, system-level insights.
- Bioinformatics serves as the core computational framework for processing, analyzing, and interpreting high-dimensional omics datasets using statistical models, machine learning, and network-based methods.
- The integration of multi-omics and bioinformatics enables advanced applications

Popular Biological Databases & Tools

- GenBank
- BLAST (Basic Local Alignment Search Tool)
- PDB (Protein Data Bank)
- Swiss-Prot
- Uni-Prot

NCBI (National Center for Biotechnology Information)

- A **public biological database platform** maintained by the **U.S. National Institute of Health (NIH)**
- Provides free access to DNA, RNA, protein, and genome data
- Major resources: GenBank, Protein, Genome, PubMed, BLAST, Taxonomy, SRA
- Widely used for sequence analysis, organism identification, comparative genomics, and literature search
- Plays a key role in modern microbiology, bioinformatics, and biomedical research

ncbi.nlm.nih.gov

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

All Databases Search

NCBI Home
Resource List (A-Z)
 All Resources
 Chemicals & Bioassays
 Data & Software
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 Domains & Structures
 Genes & Expression
 Genetics & Medicine
 Genomes & Maps
 Homology
 Literature
 Proteins
 Sequence Analysis
 Taxonomy
 Training & Tutorials
 Variation

All Databases
 Assembly
 Biocollections
 BioProject
 BioSample
 Books
 ClinVar
 Conserved Domains
 dbGaP
 dbVar
 Gene
 Genome
 GEO DataSets
 GEO Profiles
 GTR
 Identical Protein Groups
 MedGen
 MeSH
 NLM Catalog
 Nucleotide

NCBI
 The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.
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[PubChem](#)

NCBI News & Blog
 GenBank Release 269.0
 22 Dec 2025
 GenBank release 269.0 (12/16/2025) is now available on the NCBI FTP site. This release has 49 73 trillion bases and 6 03
 Upcoming Change in My Bibliography
 08 Sep 2025
 Use of earliest publication date for determining NIH Public Access compliance As shared in the July
 An Updated Bacterial and Archaeal

<https://www.ncbi.nlm.nih.gov/>

More Databases of Nucleotide Sequence

- DDBJ = DNA Data Bank of Japan
- ENA = European Nucleotide Archive

The screenshot shows the ENA website interface. At the top, there's a navigation bar with links: EMBL-EBI home, Services, Research, Training, About us, and EMBL-EBI. Below this is the ENA logo and the text 'European Nucleotide Archive'. There are two search input fields: 'Enter text search terms' with a search button and 'Enter accession' with a 'View' button. Below the search fields, there's a navigation bar with links: Home, Submit, Search, About, and Support. A message states: 'We recommend that you subscribe to the ENA-announce mailing list for updates on ENA services.' Below this, the title 'European Nucleotide Archive' is followed by a description: 'The European Nucleotide Archive (ENA) provides a comprehensive record of the world's nucleotide sequencing information, covering raw sequencing data, sequence assembly information and functional annotation. More about ENA.' Below the description, it says: 'Access to ENA data is provided through the browser, through search tools, through large scale file download and through the API.' There are three large buttons: 'Submit', 'Search', and 'Support'. At the bottom, there's a section for 'Latest ENA news' with two news items: 'New project to improve understanding of gene regulation landscape Dec 4, 2025, 2:48:19 PM' and 'A new gateway to global antimicrobial resistance data Nov 18, 2025, 3:44:56 PM'.

<https://www.ebi.ac.uk/ena/browser/home>

<https://www.ddbj.nig.ac.jp/index-e.html>

The screenshot shows the DDBJ website interface. At the top, there's a navigation bar with links: DDBJ, Services, SuperComputer, Statistics, Activities, About Us, and DDBJ Web Sites. Below this is the DDBJ logo and the text 'Bioinformation and DDBJ Center provides sharing and analysis services for data from life science researches and advances science.' There are six large buttons: 'Search', 'Submission', 'Services', 'Super Computer', 'Statistics', and 'Activities'. At the bottom, there's a section for 'About us'. On the right side, there's a 'NEWS' section with the title 'Suspension of DDBJ Center activity during the New Year Holidays' and the text 'WGS entries from ENA during a specific period are not being made available.' Below the news section, there's a 'more' link.

GenBank

 An official website of the United States government [Here's how you know](#) ▼



Log in

GenBank

Nucleotide ▼

Search

GenBank ▼

Submit ▼

Genomes ▼

WGS ▼

Metagenomes ▼

TPA ▼

TSA ▼

INSDC ▼

Documentation ▼

Other ▼

About GenBank

GenBank Data Usage

The GenBank database is designed to provide and encourage access within the scientific community to the most up to date and comprehensive DNA sequence information. Therefore, NCBI places no restrictions on the use or distribution of the GenBank data. However, some submitters may claim patent, copyright, or other intellectual property rights in all or a portion of the data they have submitted. NCBI is not in a position to assess the validity of such claims, and therefore cannot provide comment or unrestricted permission concerning the use, copying, or distribution of the information contained in GenBank.

Confidentiality

Some authors are concerned that the appearance of their data in GenBank prior to publication will compromise their work. GenBank will, upon request, withhold release of new submissions for a specified period of time. A date must be specified; we can not hold a sequence indefinitely pending publication. However, if the accession number or sequence data appears in print or online prior to the specified date, the sequence will be released. In order to prevent the delay in the appearance of published sequence data, we urge authors to inform us of the appearance of the published data. As soon as it is available, please send the full publication data--all authors, title, journal, volume, pages and date--to the following address: update@ncbi.nlm.nih.gov

Privacy

If you are submitting human sequences to GenBank, do not include any data that could reveal the personal identity of the source. GenBank assumes that the submitter has received any necessary informed consent authorizations required prior to submitting sequences.

[Disclaimer](#)

[Privacy statement](#)

Access to GenBank

There are several ways to [search and retrieve](#) data from GenBank.

[Entrez Nucleotide](#) is a collection of sequences from several sources, including GenBank. Text searches can retrieve data by searching for gene or product names, organism or author searches.

Search and align GenBank sequences to a query sequence using [BLAST](#) (Basic Local Alignment Search Tool). BLAST searches Nucleotide, dbEST, and dbGSS independently; see [BLAST info](#) for

GenBank

- A comprehensive public database of DNA and RNA sequences.
- Maintained by the National Center for Biotechnology Information (NCBI)
- Contains specific gene sequences, whole genomes, plasmid sequences.
- Sequences are submitted by researchers worldwide and assigned unique accession numbers.
- Widely used for sequence retrieval, annotation, BLAST analysis, and comparative genomics.

Accession Number

A GenBank accession number is a unique, permanent alphanumeric identifier assigned to each sequence record in GenBank, allowing it to be precisely retrieved and referenced.

GI Number: A **GI number** is a unique, sequentially assigned numeric identifier given by NCBI to each specific version of a nucleotide or protein sequence record, where any change in the sequence results in assignment of a new GI number.

GenBank Send to: Change region shown

Pseudomonas aeruginosa RRA10 blaNDM-1 gene for metallo-beta-lactamase NDM-1, partial sequence

GenBank: LC723945.1
[FASTA](#) [Graphics](#)

Go to:

LOCUS	LC723945	604 bp	DNA	linear	BCT 29-SEP-2022
DEFINITION	Pseudomonas aeruginosa RRA10 blaNDM-1 gene for metallo-beta-lactamase NDM-1, partial sequence.				
ACCESSION	LC723945				
VERSION	LC723945.1				
KEYWORDS	.				
SOURCE	Pseudomonas aeruginosa				
ORGANISM	Pseudomonas aeruginosa Bacteria; Pseudomonadati; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas.				
REFERENCE	1				
AUTHORS	Aswad,R.R., Atta,M.M. and Zamani,Z.				
TITLE	Sequencing of Carbapenem drug resistant Pseudomonas aeruginosa isolated from Patients admitted to Al-Refaae General Hospital				
JOURNAL	Unpublished				
REFERENCE	2 (bases 1 to 604)				
AUTHORS	Aswad,R.R.				
TITLE	Direct Submission				
JOURNAL	Submitted (20-AUG-2022) Contact:Reyam Rashid Aswad Azad University, Science and investigations; Tehran, Tehran 12000, Iran URL :https://srb.iau.ir/en				
FEATURES	Location/Qualifiers				
source	1..604 /organism="Pseudomonas aeruginosa" /mol_type="genomic DNA" /strain="RRA10"				

Analyze this sequence

Run BLAST

Pick Primers

Highlight Sequence Features

Find in this Sequence

Related information

Taxonomy

Recent activity

Turn Off Clear

Pseudomonas aeruginosa RRA10 blaNDM-1 gene for metallo-beta-lactamase ND Nucleotide

Pseudomonas aeruginosa NDM-1 sequence (3369) Nucleotide

Providencia stuartii strain dmpo_s979a, whole genome shotgun sequencing Nucleotide

Klebsiella pneumoniae strain BF1, whole genome shotgun sequencing project Nucleotide

Providencia stuartii strain ATCC 29914 16S ribosomal RNA gene, partial sequen Nucleotide

Downloading FASTA File from GenBank

- A sequence in FASTA format begins with a single-line description, followed by lines of sequence data.
- The description line (define) is distinguished from the sequence data by a greater-than (">") symbol at the beginning.

```
>gi|2311274183|dbj|LC723945.1| Pseudomonas aeruginosa RRA10 blaNDM-1 gene for metallo-beta-lactamase NDM-1, partial sequence
TGGACTGGCGACCAACGGTTTGGCGATCTGGTTTTCCGCCAGCTCGCACC GAATGTCTGGCAGCACACTT
CCTATCTCGACATGCCGGGTTTCGGGGCAGTCGCTTCCAACGGTTTGATCGTCAGGGATGGCGGCCGCGT
GCTGGTGGTCGATAACCGCTGGACCGATGACCAGACCGCCAGATCCTCAACTGGATCAAGCAGGAGATC
AACCTGCCGGTCGCGCTGGCGGTGGTGACTCACGCGCATCAGGACAAGATGGGCGGTATGGACGCGCTGC
ATGCGGCGGGGATTGCGACTTATGCCAATGCGTTGTGCAACCAGCTTGCCCCGCAAGAGGGGATGGTTGC
GGCGCAACACAGCCTGACTTTCGCCGCAATGGCTGGGTGCAACCAGCAACCGCGCCCACTTTGGCCCG
CTCAAGGTATTTTACCCCGGCCCCGGCCACACCAGTGACAATATCACCGTTGGGATCGACGGCACCGACA
TCGCTTTTGGTGGCTGCCTGATCAAGGACAGCAAGGCCAAGTCGCTCGGCAATCTCGGTGATGCCGACAC
TGAGCACTACGCCGCGTCAGCGCGCGCGTTTGGTGCGGCGTTCC
```

BLAST

- A sequence comparison tool developed by National Center for Biotechnology Information (NCBI)
- Used to identify sequence similarity between a query DNA/protein and database sequences
- Helps in organism identification, gene annotation, and homology analysis
- Essential tool in genomics, microbiology, and bioinformatics research

BLAST® » blastn suite

Home Recent Results Saved Strategies Help

blastn blastp blastx tblastn tblastx

Standard Nucleotide BLAST

BLASTN programs search nucleotide databases using a nucleotide query. more...

Reset page Bookmark

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ? Clear

Query subrange ?

From

To

Or, upload file

Choose File No file chosen ?

Job Title

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database

☒ Standard databases (nr etc.): ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases

Core nucleotide database (core_nt) ?

Limit SRA by

☐ Run ☐ Experiment [Extended SRA lookup](#)

Organism

Optional

Enter organism name or id—completions will be suggested

☐ exclude [Add Organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown ?

Exclude

Optional

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to

Optional

Sequences from type material

Entrez® Query

Optional

Enter an Entrez query to limit search ?

[YouTube](#) [Create custom database](#)

Program Selection

Different Types of BLAST

Nucleotide blast

Search a **nucleotide** database using a **nucleotide** query

Algorithms: blastn, megablast, discontinuous megablast

Protein blast

Search **protein** database using a **protein** query

Algorithms: blastp, psi-blast, phi-blast, delta-blast

Blastx

Search **protein** database using a **translated nucleotide** query

Tblastn

Search **translated nucleotide** database using a **protein** query

tblastx

Search **translated nucleotide** database using a **translated nucleotide** query

BLAST Outcomes

- ✚ **Max Score**- highest alignment score between query (my search) and subject/hit (database sequence).
- ✚ **Total Score**- Sum of alignment scores of all segments from the same subject that match with query.
- ✚ **Query coverage**- percent of query length that is included in the aligned segments.
- ✚ **Identity**- percent of query that is similar of subject in between the query length.
- ✚ **E- value (Expect value)**- number alignments expected by chance having a particular or better score.

Protein Data Bank (PDB)

- A central, open-access archive of three-dimensional structures of **proteins, DNA, and RNA**
- Founded in 1971 to collect, preserve, and distribute experimentally solved biomolecular structures worldwide
- Originally established at Brookhaven National Laboratory, USA
- Structural data are obtained using X-ray crystallography, NMR spectroscopy, and cryo-EM
- Each structure is assigned a unique PDB identifier (PDB ID)

<https://www.youtube.com/watch?v=cj62KET39yk>

Uni-Prot

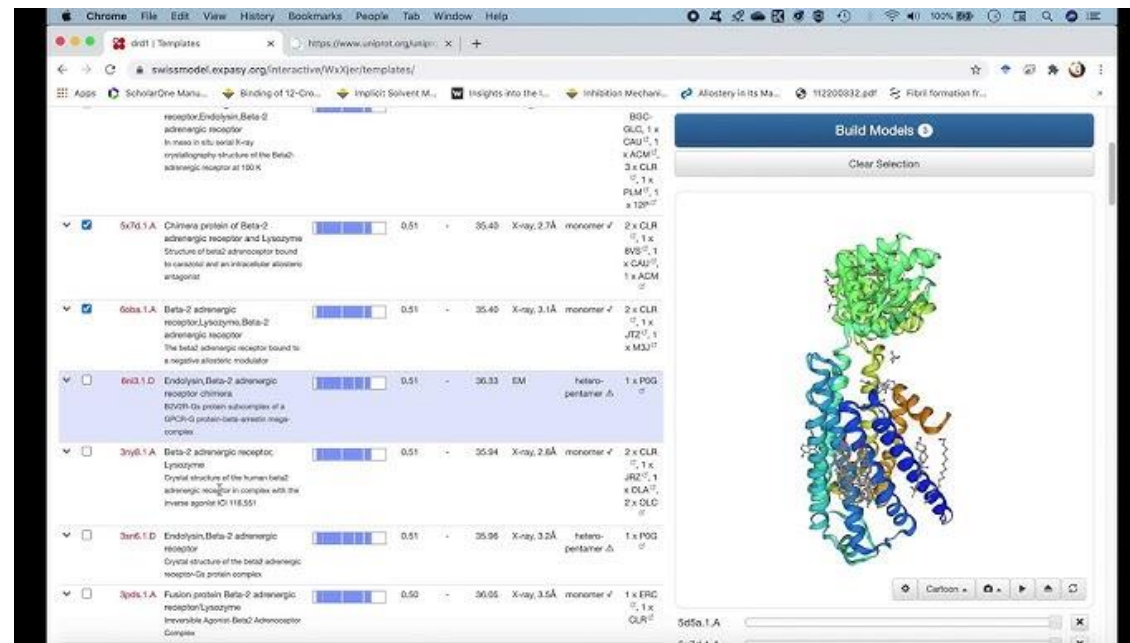
- A comprehensive, high-quality protein sequence and annotation database
 - Created to provide a centralized resource for protein sequences, functions, and biological roles
 - Contains information on protein function, domains, pathways, and post-translational modifications
 - Divided into UniProtKB/Swiss-Prot (reviewed) and UniProtKB/TrEMBL (unreviewed) sections
 - Each protein entry is assigned a unique UniProt accession number
- Maintained by the UniProt Consortium

Difference between UniProt and PDB

Feature	UniProt	Protein Data Bank (PDB)
Main focus	Protein sequence & functional annotation	3D structural information
Data type	Amino acid sequences, domains, pathways	Atomic-level 3D structures
What it explains	What the protein does	How the protein looks in 3D
Experimental data required	Not mandatory	Mandatory (X-ray, NMR, cryo-EM)
Identifier	UniProt Accession Number	PDB ID
Major applications	Functional analysis, annotation, bioinformatics	Structural biology, drug design
Level of analysis	Sequence-level & biological role	Structure-level & molecular interactions

SWISS-Model

- A web-based automated protein structure homology modeling server used to predict 3D structures of proteins when experimental structures are unavailable
- Builds models based on sequence similarity with known structures from the PDB
- Widely applied in structural bioinformatics, functional analysis, and drug discovery
- Provides quality assessment scores to evaluate model reliability

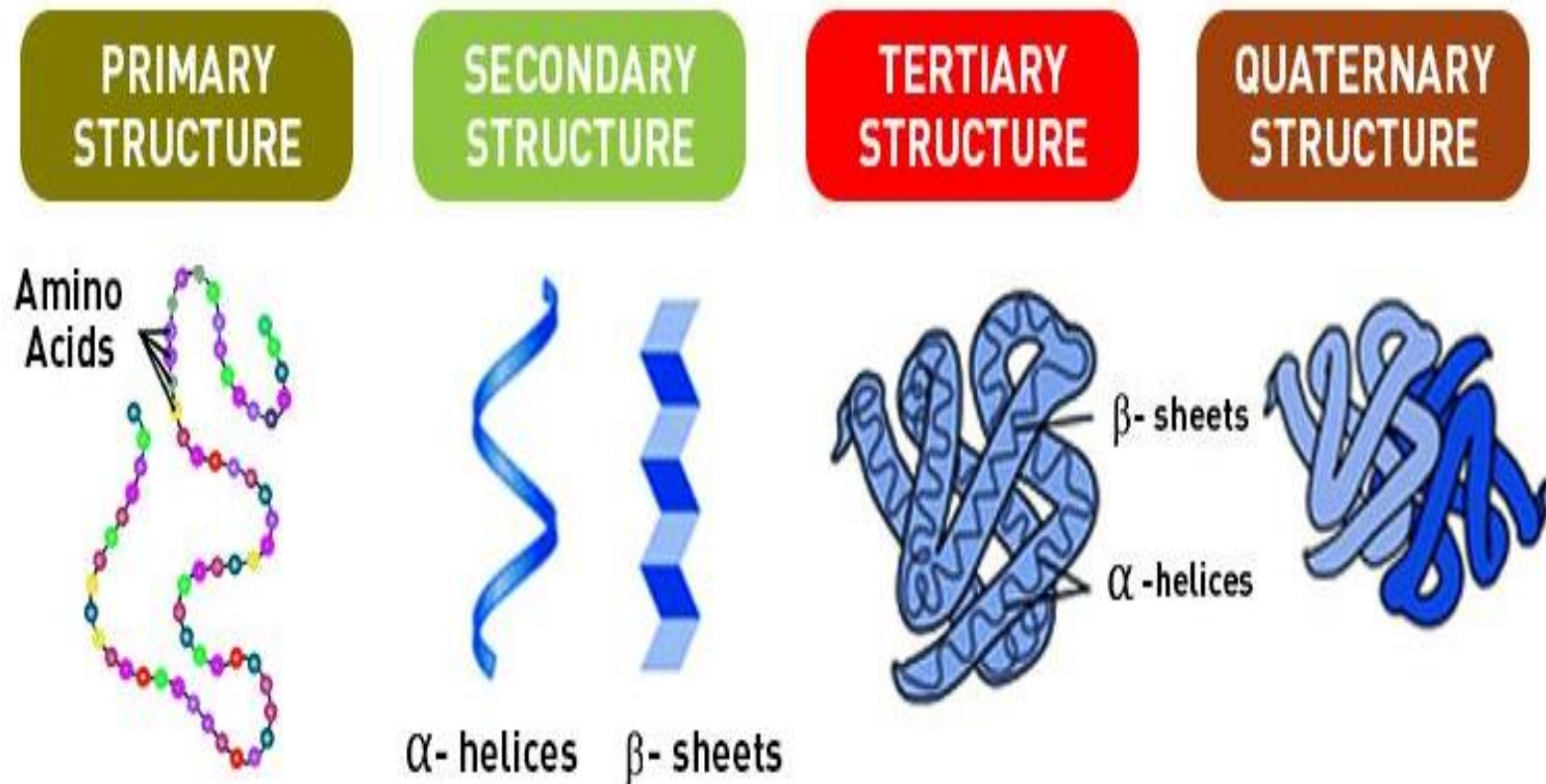


Short tutorial: <https://www.youtube.com/watch?v=k3UEwbarhPw>

Detailed tutorial: <https://www.youtube.com/playlist?list=PLoCxWrRWjqB0gLEJJx-ahPAo1278mRvCa>

Brief Idea on the Types of Protein Structures

- **Primary structure** : Linear sequence of amino acids linked by peptide bonds
- **Secondary structure**: Local folding patterns such as α -helix and β -sheet, stabilized by hydrogen bonds
- **Tertiary structure**: Overall three-dimensional (3D) structure of a single polypeptide chain
- **Quaternary structure**: Association of two or more polypeptide chains to form a functional protein complex



ML Based Tools in Proteomics: AlphaFold 3

- AlphaFold 3 is an advanced deep learning model developed for biomolecular structure prediction
- Extends previous AlphaFold versions beyond proteins to protein–protein, protein–DNA, protein–RNA, and protein–ligand complexes
- Uses transformer-based neural networks inspired by large language model (LLM) architectures
- Predicts high-accuracy 3D structures directly from sequence and molecular input

<https://blog.google/innovation-and-ai/products/google-deepmind-isomorphic-alphafold-3-ai-model/#drug-discovery>

Applications

Applications of Biological Databases and Computational Biology Tools

- **Phylogenetic Inference:** Infers evolutionary relationships and common ancestry among organisms through constructing a phylogenetic tree
- **Comparative Genomics:** Compares multiple genomes to identify shared, unique, and adaptive genes
- **AMR Profiling:** Detects antibiotic resistance genes and predicts resistance mechanisms
- **Virulence Analysis:** Identifies virulence factors associated with pathogenicity
- **Mobile Genetic Element Analysis:** Detects plasmids, integrons, transposons, and insertion sequences
- **Functional Annotation:** Assigns biological functions to genes and proteins

Applications

- **Pathway Reconstruction:** Maps genes to metabolic and cellular pathways
- **Population Genomics:** Analyzes SNPs to study strain diversity and evolution
- **Pan-genome Analysis:** Determines core and accessory gene composition of a species
- **Metagenomic Profiling:** Characterizes microbial communities from environmental samples
- **Protein Structure Prediction:** Predicts 3D structures to infer protein function
- **Drug Target Identification:** Identifies conserved genes suitable for therapeutic targeting

Tools for WGS Analysis: Galaxy Server

- Galaxy is an open-source, web-based platform for data-intensive biomedical and bioinformatics research
- Enables complex bioinformatics analyses without programming skills
- All analyses are performed directly through a web browser
- **usegalaxy.org** is a public Galaxy server freely accessible worldwide
- Provides access to thousands of integrated bioinformatics tools
- Ensures reproducible research by automatically recording complete analysis histories
- Supports workflow creation to automate multi-step analyses
- Handles major genomic file formats (FASTA, FASTQ, BAM, VCF)
- Powered by high-performance computing infrastructure
- There are 3 versions of Galaxy Server- Galaxy, Galaxy Europe, Galaxy Australia

Tools for WGS Analysis: Galaxy Server

Galaxy is an open source, web-based platform for data intensive biomedical research. If you are new to Galaxy start here or consult our help resources. You can install your own Galaxy by following the tutorial and choose from thousands of tools from the Tool Shed.

Galaxy has no border

Donate to the James P. Taylor Foundation for Open Science

Learn More

Galaxy version 25.1.rc1, commit c79e53db082ac8cf946a08c9a8274bada06c45a5

Galaxy Europe

Tools

AnyLabeling Interactive Effortless data labeling with AI support from YOLO and Segment Anything!

Epifany Runs a Bayesian protein inference

Canyon-8 estimation of ocean CO2 variables and nutrient concentrations

Filter data on any column using simple expressions

Summary Statistics for any numerical column

EMBOSS: needle Many-to-many Needleman-Wunsch global alignment

snippy-clean_full_align Replace any non-standard sequence characters in snippy 'core.full.align' file.

QualityControl Computes various QC metrics from many possible input files (only the consensusXML is required). The more optional files you provide, the more metrics you get.

QCShrinker This application is used to remove the verbose table attachments from a qcml file that are not needed anymore, e.g. for a final report

Calculate beta diversity (pairwise sample dissimilarity) on one or manyotu tables (beta_diversity)

Visualize with Krona Visualise any hierarchical data

MBioMod Reheader takes a BAM file and

There is an InteractiveTool result view available. Open

You may also access all active InteractiveTools from the User menu.

Started tool AnyLabeling Interactive and successfully added 1 job to the queue.

It produces this output:

- 25: AnyLabeling Interactive on data 22, data 21, and others: version.txt

You can check the status of queued jobs and view the resulting data by refreshing the History panel. When the job has been run the status will change from 'running' to 'Finished' if completed successfully or 'error' if problems were encountered.

Galactic News

Galaxy Europe News

This page contains announcements of interest to the Galaxy Community. These include items from the Galaxy Team or the Galaxy community and address anything that is of wide interest to the community.

Also see the Galactic Blog for more.

Contribute

To add a news item, create a new folder within the news folder with an index.md file. It's probably easiest to copy the format from one of the other existing news items as a template.

23 May 2025 Training Infrastructure as a service (TaaS) Feedback from Dr. Janne M. Tolonen Thanks to TaaS, our RNA-Seq workshops run smoothly—no more delays, just effective hands-on learning.

20 May 2025 Exploring data from BERD Data Portal with Galaxy Find out how to explore your Data from the NFDI BERD Data Portal directly in Galaxy

18 May 2025 Training Infrastructure Feedback from the From sequencing to R scripting—TaaS helped us turn RNA-Seq into a

Galaxy | Australia

Workflow Visualize Data Help Log In or Register Using 0%

Tools

FILE AND META TOOLS

Get Data

Send Data

Collection Operations

GENERAL TEXT TOOLS

Text Manipulation

Filter and Sort

Join, Subtract and Group

GENOMIC FILE MANIPULATION

FASTA/FASTQ

FASTQ Quality Control

SAM/BAM

BED

VCF/BCF

Nanopore

Galaxy Australia has been upgraded to release 24.1. Check the release notes for...

Home News People About Support Docs

Dude where's my ear tools?

They're now in the new activity bar on the left of your screen!

Get rapid, one click access to tools and other common functions:

Workflow Visualisation Histories History Multiview Datasets Pages

Galaxy Australia is an open, web-based platform for accessible, reproducible and transparent computational research. Galaxy supports thousands of documented and maintained tools that are

Commonly Used Tools of Galaxy Server

- **FastQC:** Evaluates the quality of raw sequencing reads
- **SPAdes:** Assembles sequencing reads into contigs or draft genomes
- **QUAST:** Assesses genome assembly quality and completeness
- **Prokka:** Performs rapid annotation of prokaryotic genomes
- **ABRicate:** Screens genomes for antimicrobial resistance and virulence genes along with their start and end position
- **RGI (CARD):** Identifies antibiotic resistance genes using the CARD database

Commonly Used Tools of Galaxy Server

- **PlasmidFinder:** Detects plasmid replicons in bacterial genomes\
- **MOB-suite:** Predicts plasmid-associated sequences and mobility
- **IntegronFinder:** Identifies integrons, gene cassettes, and integrase genes
- **ISEScan:** Detects insertion sequences and transposable elements
- **MLST:** Determines sequence types for bacterial strain typing
- **Roary:** Performs pan-genome analysis across multiple bacterial genomes

Construction of Phylogenetic Tree

1. Find and download the sequences to be included in the tree

NCBI

2. Align the acquired sequences, check and trim the alignment

Crustal W/X

MEGA 5/6/7

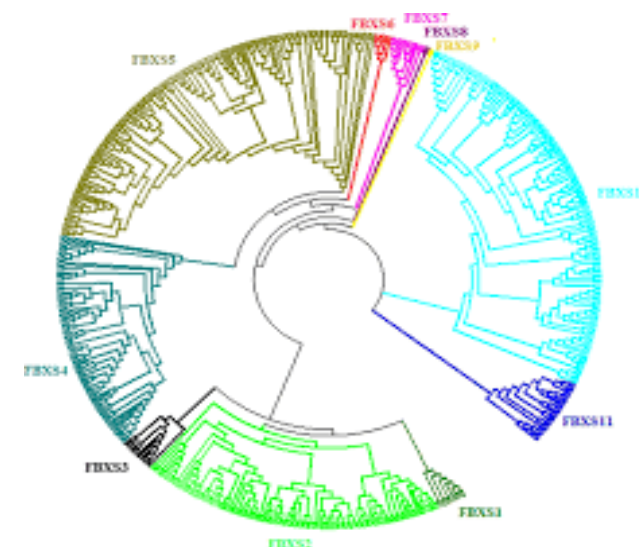
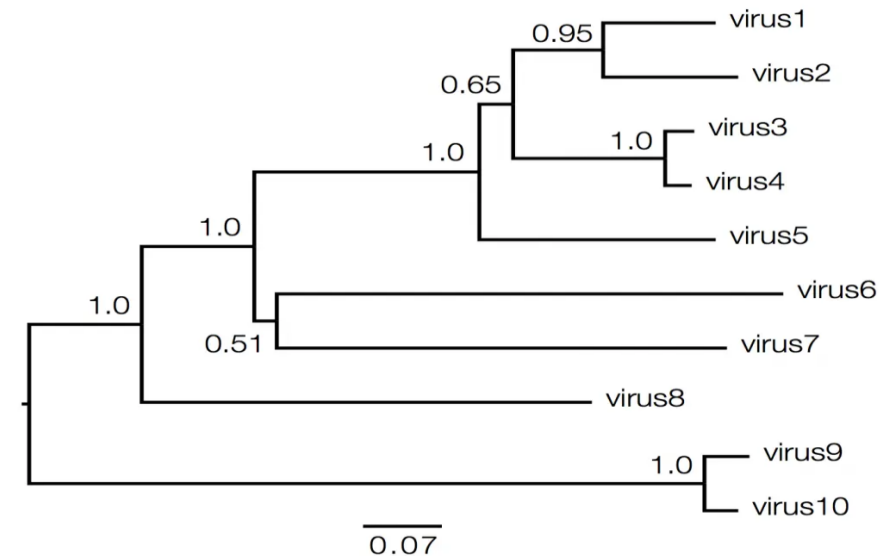
3. Construct the phylogenetic tree

MEGA 5/6/7

4. Visualization of phylogenetic tree

iTOL, FigTree

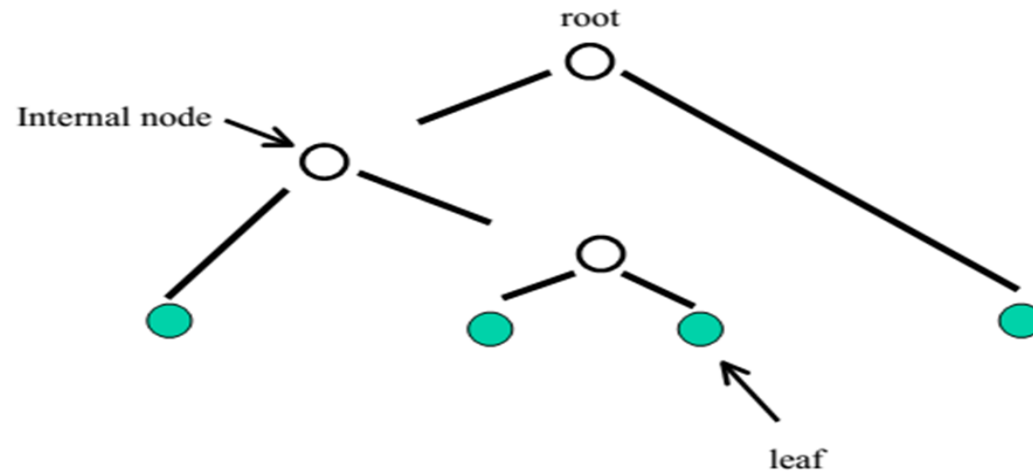
<https://www.youtube.com/watch?v=ry95ObEBK4>



Concept of Phylogenetic Tree

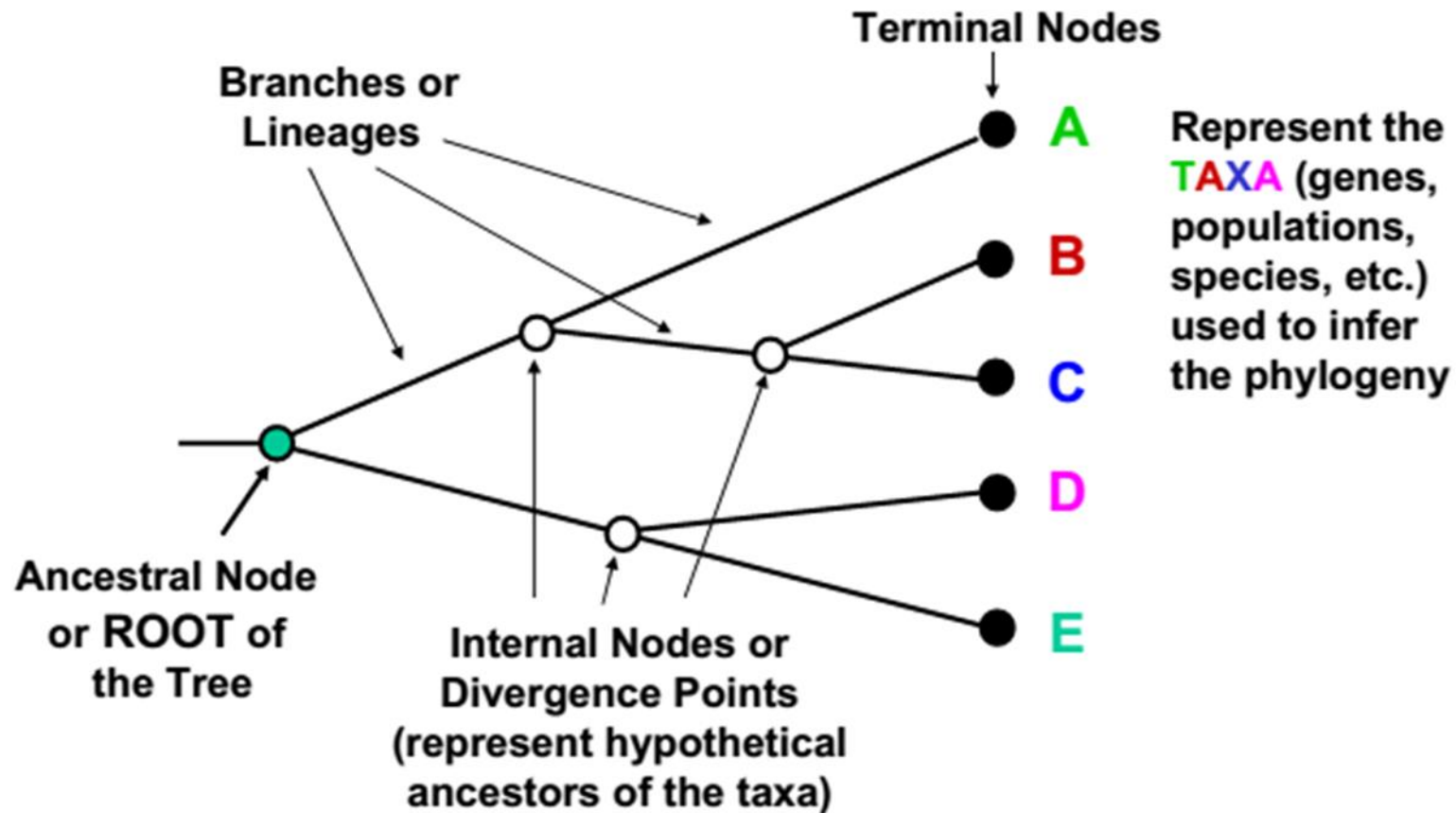
A phylogenetic tree (evolutionary tree) is the graphical representation of the evolutionary history of biological sequences and allows us to visualize the evolutionary relationships between them.

Tree terminology



Concept of Phylogenetic Tree

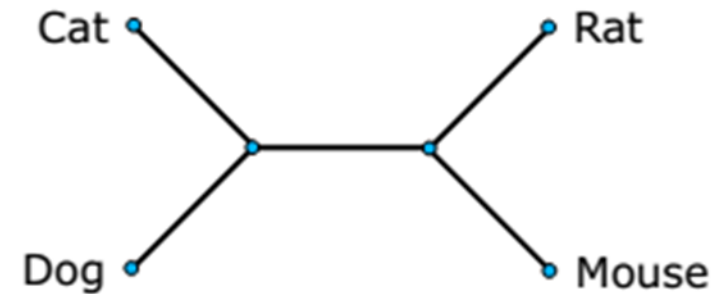
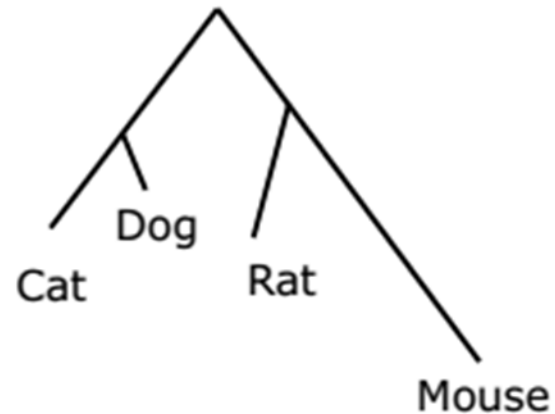
Common Phylogenetic Tree Terminology



Concept of Phylogenetic Tree

Tree basics

- Trees can be weighted or unweighted

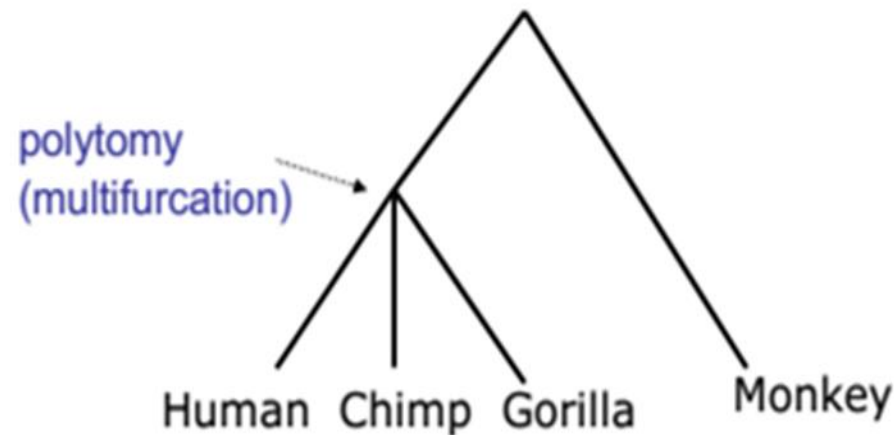


- Edge weights are used to represent
 - the amount of genetic change along the edge
 - time

Concept of Phylogenetic Tree

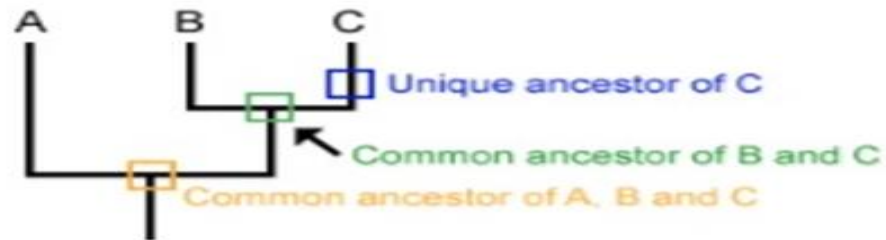
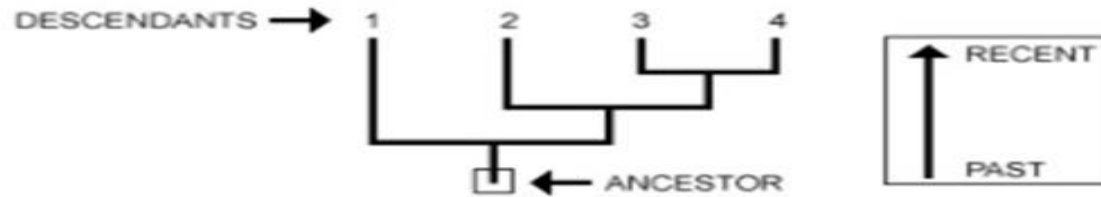
Tree basics

- Trees can be bifurcating (binary) or multifurcating (non-binary)

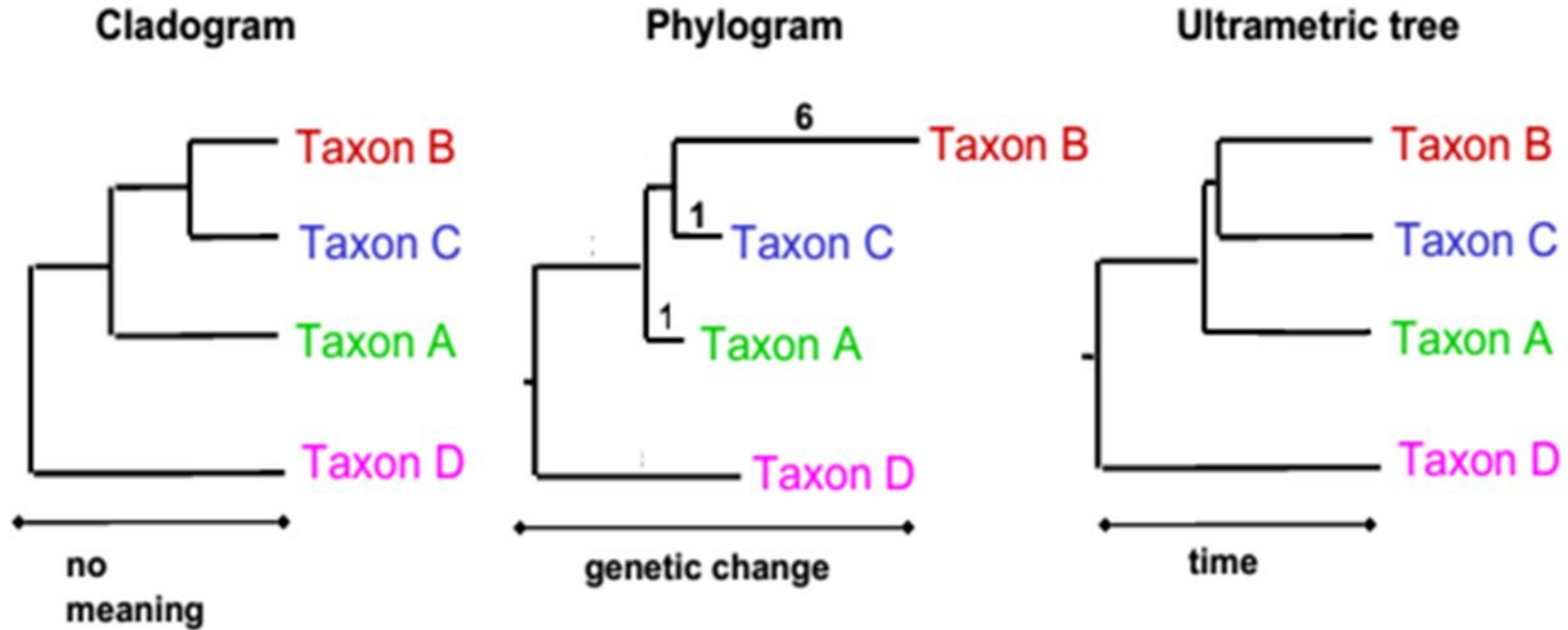


Polytomies usually represent uncertainty (*a soft polytomy*), but sometimes they are used to mean that everything happened at exactly the same time (*a hard polytomy*).

Understanding Phylogenies



Types of Phylogenetic Tree



Types of Phylogenetic Tree

- A **cladogram** shows only the branching pattern of taxa based on shared ancestry without considering branch length.
- A **phylogram** represents evolutionary relationships where branch length reflects the amount of genetic change.
- An **ultrametric tree** displays evolutionary relationships scaled to time with all taxa equidistant from the root.

Applications of Phylogenetic Tree

- Phylogenetic trees can be used to study the evolutionary relationships between different species and to understand the evolutionary processes over time.
- Phylogenetic trees can be used to study the diversity and distribution of species and to develop conservation strategies to protect endangered species and ecosystems.
- Phylogenetic trees can be used to identify the origins of pathogens and to track the spread of diseases.
- Phylogenetic trees can also be used in forensics to identify the origins of biological samples found at crime scenes and to link suspects to crimes.
- Phylogenetic trees are useful for organizing and classifying organisms and species according to their DNA sequences and morphological similarities and differences.

Thank You